

Designing LLM Response Layouts for XR Workspaces in Vehicles

Daun Kim

Seoul National University of Science and Technology
South Korea
daun@seoultech.ac.kr

Jin-Woo Jeong*

Seoul National University of Science and Technology
South Korea
jinw.jeong@seoultech.ac.kr



Figure 1: Examples of LLM response layouts as viewed by participants in Vehicle XR Environments

Abstract

This study investigates the design of response layouts for large language models in XR environments for vehicle settings. Our formative study identified usability challenges with the current linear layout, such as difficulties in extensive scrolling and mid-air interactions. Based on these findings, we designed an improved layout that utilizes multiple windows for efficient access to diverse content, with a cohesive placement to minimize motion sickness. A user study with 24 participants compared the proposed interface against a conventional web browser interface through various information-seeking tasks. The results demonstrated that the proposed interface enhanced overall usability compared to the baseline.

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1 Introduction

In this study, we explore user interactions and perceptions of multimodal Large Language Model (LLM) responses through extended reality (XR) devices in vehicles. Recently, as people spend more time traveling, they increasingly seek ways to be more productive in vehicles [Li et al. 2021]. With this context, integrating LLMs with in-vehicle XR is expected to introduce a unique and promising workspace, but presents both unique challenges as well. For

example, to harness the power of LLMs on XR devices, users need to utilize the built-in web browsers, as illustrated in Fig.1-Baseline. However, contents delivered through web browsers are typically optimized for desktop environments [Xing et al. 2022]. Moreover, vehicle-specific issues like limited space, restricted body movement, and motion vibrations add further complications. Despite the importance and potential of their intersection, research on the application of LLMs within in-vehicle XR environments is still limited. Our study aims to fill this gap by investigating how different layouts of LLM responses affect the user's in-vehicle XR experiences. To this end, we identify the challenges in traditional LLM response layout, and then explore the optimal layout design of LLMs for in-vehicle XR experiences, with a consideration of various factors like information placement and interaction ease, while also addressing vehicle-specific challenges in XR workspace design.

2 System design

We conducted a formative study with 18 participants to assess perceptions of LLMs in vehicular XR environments using Microsoft HoloLens 2. Participants interacted with the Gemini interface, followed by interviews to gather feedback and identify areas for improvement. Participants then used a custom Unity app to redesign the LLM response layout based on their preferences and needs. As a result, we derived three design considerations: 1. **Cohesive information layout**: users preferred a cohesive layout with content arranged within a single field of view (FOV), which can minimize head movement. This is consistent with previous findings that restricting the display size within the FOV improves task efficiency. Also, participants preferred side-placed content angled towards them, which mirrors the benefit of curved displays that provide equidistant viewing and improve visibility and accessibility [Mcgill et al. 2020]. 2. **Low-collision risk space**: participants expressed concerns about potential collisions in the constrained vehicle space. Design emphasis should focus on the more spacious central region

*Corresponding author

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of the vehicle rather than the constrained areas near the windows.

3. Multi-Window Presentation of Segregated Modalities: participants reported arm fatigue from prolonged mid-air scrolling and difficulties with information overload on a single webpage, which leads to errors like unintentional modality selection (e.g., trying to zoom in on a map but unintentionally dragging text). Participants suggested separating different modalities into distinct windows. Specifically, participants preferred placing interactive and exploratory elements, such as text and maps, centrally, while locating quickly scanned items like tables, videos, and images toward the periphery. Based on these design considerations, we developed an enhanced layout for LLMs as illustrated in Fig.1-Our system. For implementation, Unity 2022.3.1f1. and Mixed Reality Toolkit v2.8.3. were used. More details available at video.

3 User study

We recruited 24 participants (13 males, 11 females) aged 21-28 years ($M=24.17$, $SD=1.97$) from a local university. The study was conducted on a low-traffic, closed-loop oval circuit test track at the university, using a Chevrolet Malibu driven by a trained researcher at an average speed of 20 km/h and a maximum speed of 30 km/h. Participants were equipped with HoloLens2 and completed information-seeking (IS) tasks using Gemini with two conditions: Baseline (Built-in Web browser) and the derived response layout. For each condition, participants engaged in 6 sessions and were asked to verbally answer three questions per session. The experiment used a within-subjects design, with the conditions and IS tasks counterbalanced with a Latin square. During each condition, participants reported their level of motion sickness using MSTT scores: after the first three sessions and after the last three. They also completed NASA-TLX and System Usability Scale (SUS) questionnaires after each condition, with a 10-minute break between conditions. Also, we recorded the user's task completion time for each condition for further analysis. A semi-structured interview followed the experiment to gather qualitative feedback on both interfaces. More details are available in the supplementary material.

4 Result

The Shapiro-Wilk test showed non-normality, so we used the Wilcoxon test to compare the two visualization conditions.

NASA-TLX: As illustrated in Fig.2, our approach achieved a significantly lower overall workload ($M=6.30$, $SD=3.00$) compared to the baseline ($M=10.51$, $SD=4.51$) ($p < .001$). Notably, our method significantly reduced mental and physical demands ($p < .001$) while enhancing users' perception of performance ($p < .01$). This indicates that users experienced reduced cognitive strain and less exertion to accomplish their goals with our system.

Completion time: Participants using our design layout completed tasks significantly faster ($M=49.70$, $SD=14.43$) compared to the baseline condition ($M=75.67$, $SD=22.21$) ($p < .001$). This finding strongly emphasizes the effectiveness of our proposed design in facilitating navigation for information seeking when faced with complex LLM responses encompassing multiple modalities.

SUS: Compared to the baseline achieving an average score of 53.13, the proposed approach received a significantly higher average

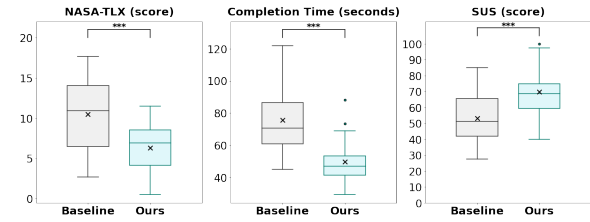


Figure 2: Result (* $p < .05$, ** $p < .01$, *** $p < .001$)

score of 69.79 ($p < .001$), demonstrating a marked improvement in usability. In particular, the proposed approach scored significantly higher in terms of willingness to use frequently ($p < .001$) and scored higher on the ease of use ($p < .01$). These results indicate that the proposed design could offer substantially enhanced usability.

Motion Sickness: MSTT results showed low motion sickness levels for both the baseline ($M=2.81$, $SD=2.03$) and our proposed method ($M=3.15$, $SD=1.99$), ranging between the level of "not noticeable" and "harmless". Also, no significant difference was found between conditions ($p = .28$). Further analysis revealed that there was a significant increase from the first to the second measurement in both the baseline condition (Baseline1: $M=2.38$, $SD=1.76$; Baseline2: $M=3.25$, $SD=2.25$; $p < .01$) and the proposed method (Ours1: $M=2.75$, $SD=1.78$; Ours2: $M=3.54$, $SD=2.19$; $p < .05$). Nevertheless, average scores remained below the level of 'uncomfortable', which means both conditions remain practical. This finding demonstrates that our approach can maintain motion sickness levels comparable to the baseline while improving usability.

5 Conclusion and future work

This study investigated user interactions with LLM responses via in-vehicle XR devices and identified usability challenges and design considerations. Our proposed design significantly improved usability over the baseline, emphasizing the importance of tailoring LLM interactions to in-vehicle XR constraints and opportunities. However, our study has several limitations: 1) The controlled setting may not reflect real driving conditions, 2) Predefined interactions may limit system exploration, and 3) Advanced modalities like gaze were not considered. Further research is also needed to understand the potential and limitations of XR workspace in automotive use.

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